

2021-Ph-H1-Q6

Section: Our Dynamic Universe

Topic: Energy and Power

A rollercoaster car of mass 200 kg falls 20 m.

What is the gain in kinetic energy?

Solution:

Energy lost as potential energy becomes kinetic energy:

$$E_k = E_p = mgh = 200 \times 9.8 \times 20 = 39\,200 \text{ J}$$

Answer: C

Guidance for Students:

In frictionless motion, energy is conserved. All gravitational potential energy becomes kinetic energy.

Revision Tip:

Learn this identity:

$$\Delta E_k = \Delta E_p = mgh$$

Use it when an object changes height.